

Incorporating Spatial Dependence

Constructing Insurable Risk Portfolios

Colombia Workshop

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First Principles

What About Spatial Dependencies?

- Think about pricing/underwriting insurance for
 - A home in Florida (subject to hurricane risks)
 - A business in California (subject to earthquake risks)
 - A farm in Iowa (subject to hail and other climate risks)

We could adjust our actions for “market conditions” but it would also be nice to offer a model that incorporates spatial dependencies using first principles. . .

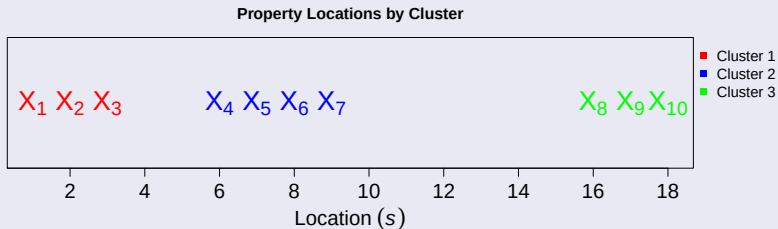
the subject of this section of the workshop.

Road Map - Spatial Dependencies

- A Toy Example
- NFIP Flood Data
- Demonstrations
 - Predicted Claims Distribution
 - Managing Exposure
 - Top-Down Approach to Pricing
 - Risk Exchanges

Incorporating Spatial Dependence - A Toy Example

- An owner has $n = 10$ properties with random losses $X_1, \dots, X_{10}$.
- Location of the properties (s) are known.
 - They fall along a coast (for now, we limit spatial considerations to a line)
 - The locations are in three clusters, as given below:



- For simplicity, assume that risks are homogeneous (same size)
- To cede a bit of risk from this portfolio, how can one account for their relative proximity?

A Linear World

- Risks are denoted as: $\mathbf{X} = X_1, \dots, X_n$
 - For the j th risk, we know the spatial location, s_j
- Moments: mean $E(\mathbf{X})$ and variance covariance matrix $\Sigma = \text{Var}(\mathbf{X})$.
 - The variance covariance matrix Σ will be determined using spatial locations $s_1, \dots, s_n$.
- For the correlation structure, let us use the dependence function

$$K(s_i, s_j) = \exp \left(-\frac{|s_i - s_j|}{\phi_{dep}} \right)$$

Spatial Correlations

- With $\phi_{dep} = 5$, this produces the following values for Σ .
 - Strong dependencies within clusters and
 - Weaker, yet still positive, dependencies between clusters.

	1	2	3	6	7	8	9	16	17	18
1	1.00	0.82	0.67	0.37	0.30	0.25	0.20	0.05	0.04	0.03
2	0.82	1.00	0.82	0.45	0.37	0.30	0.25	0.06	0.05	0.04
3	0.67	0.82	1.00	0.55	0.45	0.37	0.30	0.07	0.06	0.05
6	0.37	0.45	0.55	1.00	0.82	0.67	0.55	0.14	0.11	0.09
7	0.30	0.37	0.45	0.82	1.00	0.82	0.67	0.17	0.14	0.11
8	0.25	0.30	0.37	0.67	0.82	1.00	0.82	0.20	0.17	0.14
9	0.20	0.25	0.30	0.55	0.67	0.82	1.00	0.25	0.20	0.17
16	0.05	0.06	0.07	0.14	0.17	0.20	0.25	1.00	0.82	0.67
17	0.04	0.05	0.06	0.11	0.14	0.17	0.20	0.82	1.00	0.82
18	0.03	0.04	0.05	0.09	0.11	0.14	0.17	0.67	0.82	1.00

A Linear Retention Problem

- **Retention (reinsurance) plan** - “quota share”
 - Consider retention weights $c_1, \dots, c_n$,
 - the risk owner retains $S(c_1, \dots, c_n) = \sum_i c_i X_i = \mathbf{c}'\mathbf{X}$
 - the risk owner transfers $\sum_i (1 - c_i) X_i = (\mathbf{1} - \mathbf{c})'\mathbf{X}$
- **Summary measures**
 - Variance of retained risk, $\text{Var}(\mathbf{c}'\mathbf{X}) = \mathbf{c}'\Sigma\mathbf{c}$
 - The (fair) risk transfer cost is $RTC(\mathbf{c}) = (\mathbf{1} - \mathbf{c})'\mathbf{E}(\mathbf{X})$.

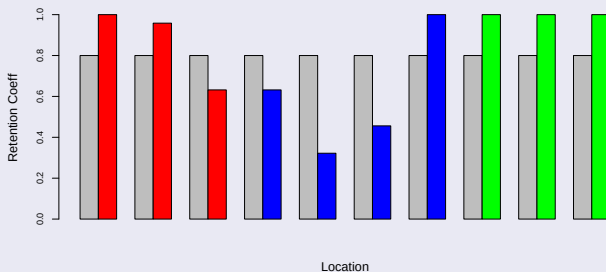
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 - The (fair) risk transfer cost is $RTC(\mathbf{c}) = (\mathbf{1} - \mathbf{c})'\mathbf{E}(\mathbf{X})$.
- **Optimization Problem** The risk owner wishes to spend at most a fixed amount, RTC_{max} , to transfer a portion of the risk. Subject to this constraint, the objective is to minimize the uncertainty of retained risks.

minimize _c	$\mathbf{c}'\Sigma\mathbf{c}$	<i>uncertainty</i>
subject to	$RTC(\mathbf{c}) \leq RTC_{max}$	<i>budget restriction</i>
	$0 \leq c_i \leq 1$	<i>parameter restrictions.</i>

Problem Solution - Dependence Influences Exposure

- The grey bars give *equal* retention rates. It is the result of assuming **independence** among risks
- The other bars represent optimal rates that account for **spatial dependence**
 - Note the clustering affect
 - The retention rates are highest for the first and last risks - they are physically the most separate and so have the lowest dependence with other risks. These are risks that the owner wishes to retain.



NFIP Flood Data

- We consider U.S. National Flood Insurance Program (NFIP) policies in the state of Florida with effective dates between January 1, 2021 and March 31, 2023, yielding approximately 1.8 million property–year observations.
- Policies with effective dates on or before July 15, 2022 are used for training, while those effective after this date form the test set.
- This time-based split results in approximately 70% of the observations in the training sample and 30% in the test sample.
- Table 1 reports summary statistics for the training data.

Table 1: Description of Data

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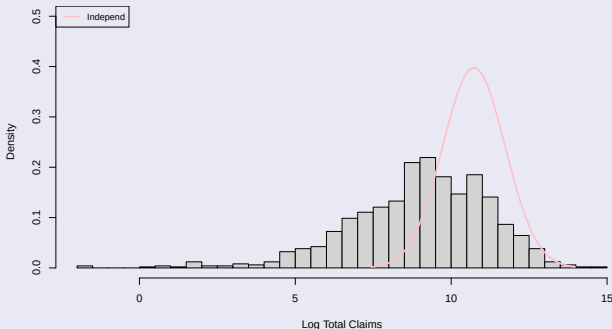
Variable	Description	Mean	SD
<i>Property Level</i>			
totalBuildingInsuranceCoverage	Building coverage	228,903	69,685
totalContentsInsuranceCoverage	Contents coverage	80,755	50,638
buildingDeduct	Building deductible	1,912	2,572
contentsDeduct	Contents deductible	1,830	2,569
yearOfBuilt	Year of built	1,989	19
numberOfFloorsInInsuredBuilding	Number of floors	1.37	0.69
basementY	= 1 if there is basement	0.77 %	
contentsHigh	= 1 if contents are above ground and higher	91.43 %	
occupancyGroup	Occupancy type		
	= 1 single family	90.69 %	
	= 2 multi-unit residential	4.65 %	
	= 3 non-residential	2.93 %	
	= 4 others (reference)		
mainHouseIndicator	= 1 if it is main house	86.07 %	
primaryResidenceIndicator	= 1 if it is primary residence	82.22 %	
tenantIndicator	= 1 if policyholder is a tenant	0.27 %	
elevatedBuildingIndicator	= 1 if it is an elevated building	8.23 %	
floodproofedIndicator	= 1 if the structure is flood proofed	0.002 %	
construction	= 1 if the building is under construction	0.07 %	
condoY	= 1 if condominium	5.15 %	
agricultureStructureIndicator	= 1 if being agricultural structure	0.009 %	
nonProfitIndicator	= 1 if being non-profit	0.36 %	
houseOfWorshipIndicator	= 1 if being a house of worship	0.04 %	
smallBusinessIndicatorBuilding	= 1 the insured represents a small business	0.79 %	
rentalPropertyIndicator	= 1 if the property is a rental property	3.34 %	
stateOwnedIndicator	= 1 if state owned	0.03 %	
ratedFloodZoneNew	need more documentation		
<i>Census Tract Level</i>			
ALAND	Land area (square kilometers)	32.92	135.73
AWATER	Water area (square kilometers)	3.69	31.11

Estimation and Prediction

- **Estimation** - We fit this model to the training data using likelihood methods.
- **Prediction** - We took a random sample of 300 census tracts and use all of the houses within the selected tracts.
 - This resulted in a sample of 26,174 policyholders.
 - For each policyholder, we generated 1,000 simulated losses
- For comparison purposes, we also simulated losses assuming
 - **Independence**
 - Setting each dependence parameter to 0 (one for τ_1 , τ_2 , and λ)
 - Also set these parameters to a small, non-zero value

Demonstration. Distributions of Total Portfolio Claims

- The distribution of total claims is **very sensitive** to the dependence parameters.
 - The forecast distribution assuming *independence* among claims is much tighter.



Summaries of Total Claims by Dependency Type

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Std Dev
Base Dependence	0	2245	10049	44408	39897	2241239	125992
Independence	34440	42944	45543	45805	48514	58388	3972

- The distribution of total claims is **very sensitive** to the dependence parameters.
- This is a forecast of claims in the future - the basis of our loss reserves
- However, this demonstration is only for a one-period model.
 - Typically, our loss reserves reflect the effects of time (think loss triangles)
 - So, this demonstration only suggests the effects of spatial dependence on loss reserves, more research needed!

Demonstration. Managing Exposure in a Spatially Dependent Risk Pool Using Constrained Optimization

- How can analysts advise insurers when spatially dependent risks are “too concentrated”?
- Insurers can manage exposure by:
 - ① Simply not underwriting, or limiting their acceptances, of risks in selected areas.
 - ② Limiting the amount of risks that they take on through retention parameters such as coinsurance and upper limit parameters.
 - ③ Ceding portions of their pool to other parties such as through reinsurance agreements.
- To quantify this, we consider an approach that dates back to de Finetti (1940).
 - We adapt it to account for spatial proximity.

Managing Exposure Risk Retention Optimization Problem

- Continue with the NFIP flood insurance data.
- To begin, consider a random sample of 300 census tracts (out of 5,508), so that our test sample of 26,174 policyholders.
 - Thus, there are $K = 300$ census tracts and j denotes the policyholder within a tract.
- We consider a quota share retention function

$$g_{QS}(\theta_1, \dots, \theta_K) = \sum_{k=1}^K \theta_k \sum_{j \in N_k} X_{kj}.$$

- Here, the coinsurance retention parameter is constant over units within a tract.
- $\theta_k = 1$ means include that tract in our portfolio, $\theta_k = 0$ means exclude that tract in our portfolio

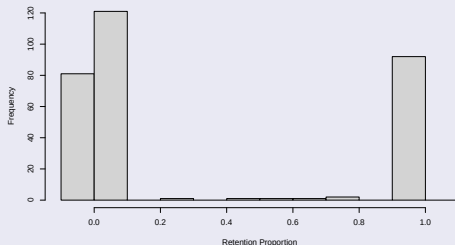
Managing Exposure Risk Retention Optimization Problem

minimize $_{\theta}$	$RM[g(\mathbf{X}; \theta)]$	<i>uncertainty</i>
subject to	$RTC(\theta) \leq RTC_{max}$	<i>budget restriction</i>
	$0 \leq \theta_j \leq 1$	<i>parameter restrictions</i>

- Instead of variance, we now use expected shortfall (also known as conditional tail expectation) for the risk measure (**RM**)
 - Like variance, its use results in a convex problem which is super fast to solve
- Use this constrained optimization to determine the optimal values of θ that indicate how much of a census tract we should expose ourselves to

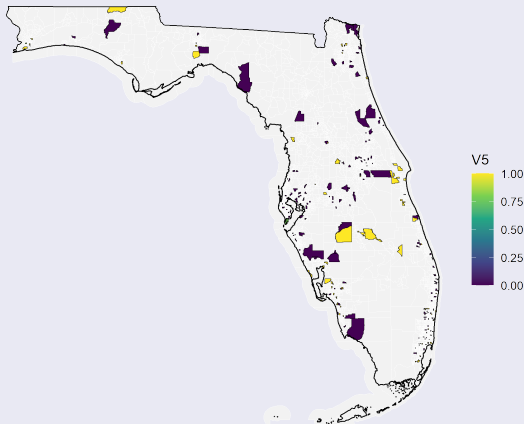
Distribution of Quota Share Retention Proportions

- This analysis is with RTC_{max} at 50% of the total cost
- Out of 300 optimal parameters:
 - Many of the retention proportions are above 0.80, suggesting full retention
 - Many are below 0.20, suggesting full transfer
 - There are a few census tracts that are in $[0.20, 0.80]$.



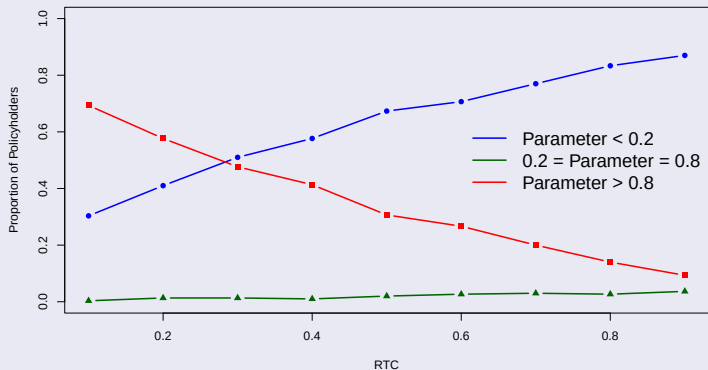
Retention Proportions 300 Census Tracts Randomly Selected

- Where do we not retain risks?
- Dark shading (indicating no retention) are near a coast

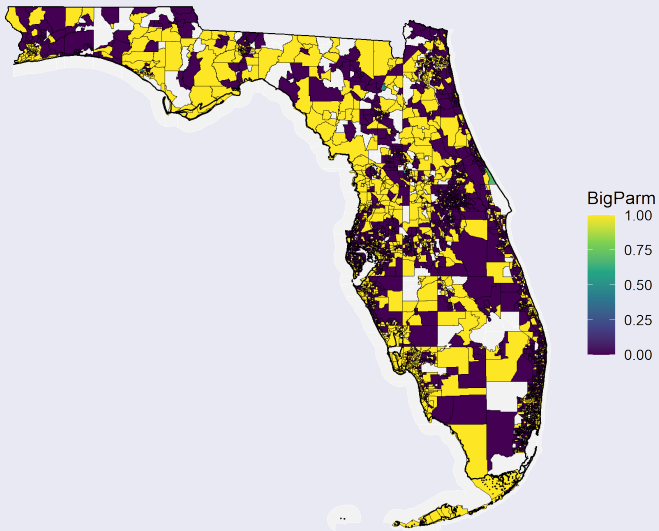


Quota Share Retention Proportions by RTC_{max}

How do the optimal retentions vary with the budget limit RTC_{max} ?



Retention Proportions - All 5,508 Census Tracts



Summary of Managing Exposure in a Spatially Dependent Risk Pool

- The constrained optimization set up allows one to provide risk retention recommendations that accounts for spatial dependence.
- Dependence matters: the optimal retention coefficients differ dramatically for different dependence structures
- The quota share optimal proportions may provide an excellent tool to help the insurer decide upon areas to focus their retention efforts.

Demonstration. Spatial Dependencies and Insurance Pricing

- To see how spatial dependencies can affect insurance pricing, let us consider the “top-down” procedure advocated by Bühlmann (1985).
 - First calculate a price for an insurance portfolio, then allocate this total premium to individual units.
- ① Determine an appropriate price for each retained risk in absence of costs of capital as well as service and administrative costs.
 - In modern-day practice, one utilizes **generalized linear models** and related statistical learning tools for individuals risks.
 - Focus on the net premium principle is $P(Z_i) = E(Z_i)$, where “E” is the expectation operator.
 - Summing the individual risks gives us a pool retained risk $(\sum_i Z_i)$ and its associated premium $(\sum_i P(Z_i))$.

Insurance Pricing

- 1 Determine an appropriate price for each retained risk in absence of costs of capital as well as service and administrative costs.
- 2 Next, one specifies the required capital for the pool retained risk. Bühlmann (1985) did this using what he referred to as a “stability criterion.”
 - For example, using a ruin criteria one may determine the required capital K as a solution of the equation

$$\alpha = \Pr \left(\sum_i Z_i > K + \sum_i P(Z_i) \right)$$

where α is an appropriately small “ruin probability.”

Insurance Pricing 2

- 1 Determine an appropriate price for each retained risk in absence of costs of capital as well as service and administrative costs.
- 2 Next, one specifies the required capital for the pool retained risk.
- 3 The total pool premium is

$$\text{Total Pool Premium} = \sum_i P(Z_i) + \text{Admin} + \text{Cost of Capital}$$

- “Admin” can represent administrative expenses and costs to service the policies (including claims)
- “Cost of Capital” is K times a required rate of return on the capital for investors.

Insurance Pricing 3

- ① Determine an appropriate price for each retained risk in absence of costs of capital as well as service and administrative costs.
- ② Next, one specifies the required capital for the pool retained risk.
- ③ Determine the total pool premium.
- ④ The total pool premium is then distributed to individual risks.
 - Part III of Mildenhall and Major (2022) describes several approaches.
 - This premium development procedure does not necessarily result in prices paid by policyholders and so are commonly referred to as “technical premiums.”
 - They can serve to provide a rate schedule that insurers may offer.

Bühlmann's Top-Down Pricing Approach

- We summarize calculations assuming spatially dependent and independent claims
 - I also show results for two types of retention levels that depend on the spatially dependence and independence.
- In sum, we learn that **the assumption of a spatial dependence on claims has an important impact on prices.**
 - This impact turns out to be robust to the type of retention level.

Bühlmann's Top-Down Pricing - NFIP Data

	Expected Cost	VaR Cost	Capital Required	Cost of Capital
Test Data Dependent	20428	150702	130275	32
Test Data Independent	21693	25187	3494	1

- Consider the top row.
 - The expected retained claims are 20428 (thousands).
 - With fair pricing, then capital required is $150702 - 20428 = 130275$.
 - If the rate of return on capital is 5%, then the cost of capital is 6514, or 32% of the expected/fair cost.
- It is not at all surprising that highly spatially dependent risks should be charged more.
 - This approach provides a **disciplined way** of assessing the extra costs due to spatial dependence.
- **Why raises prices?** A response to regulators and consumers!

Demonstration. Risk Exchange

- Reduce the uncertainty of retained risk through risk exchanges
- Imagine a very simple situation of $n = 2$ portfolios from the Florida risk data
 - Take the 5,000 policyholders in the northernmost part of the state and 1,000 policyholders in the southernmost part.
 - This results in two portfolios called “Before Exchange South” and “Before Exchange North”, respectively.

	Mean	Std Dev	VaR_{98}	ES_{98}	ES_{90}
Before Exchange South	5623	19303	52673	106858	41366
Before Exchange North	5772	12315	43547	71430	33706

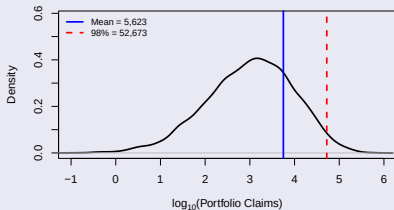
Risk Exchange Distributions

- Now let us imagine a very simple risk exchange.
 - Each portfolio, or “company,” randomly selects 50% of its portfolio and exchanges it with the other company.
 - This results in two portfolios called “After Exchange South” and “After Exchange North”, respectively.
- For each measure of volatility, the **portfolio uncertainty has reduced** for each company.

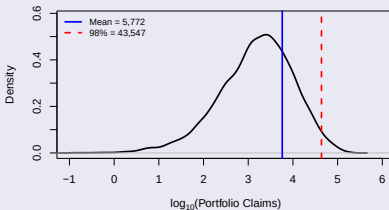
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Before Exchange South	5623	19303	52673	106858	41366
Before Exchange North	5772	12315	43547	71430	33706
After Exchange South	5717	11481	37845	64259	30257
After Exchange North	5677	11638	38390	65041	30433

Portfolio Distributions Before and After a Risk Exchange

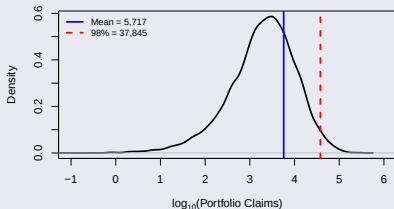
Before Exchange South



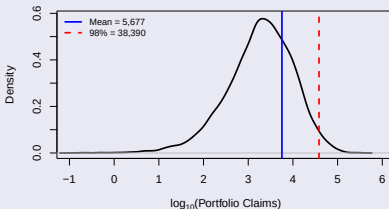
Before Exchange North



After Exchange South



After Exchange North



Some Intuition

- By encouraging companies to exchange risks with one another, they diversify their holdings.
- We used three factors to model the Florida spatial dependence.
 - One was a global shock to the entire state,
 - one that is common to all policyholders within a census tract, and
 - one that depends on the distance among policyholders.
- By exchanging a concentrated portfolio for one that is more diversified, each portfolio dramatically reduced its uncertainty.
- There is a rapidly growing academic literature on optimal risk exchanges, more research needed!

Recap

By focusing on the fundamentals, we believe that the methods described here are easily interpretable and thus more likely to be adopted in commercial practice.

- These tools include:
 - Recognition of the effects of spatial dependence on loss reserves
 - Methods of managing exposure
 - Adjustments to insurance prices
 - Reducing uncertainty of retained risk through risk exchanges
- The (academic) paper underpinning this work is available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7374878

Selected References

Bühlmann, Hans. 1985. "Premium Calculation from Top Down." *ASTIN Bulletin: The Journal of the IAA* 15 (2): 89–101.

Mildenhall, Stephen A, and John Major. 2022. *Pricing Insurance Risk: Theory and Practice*. John Wiley & Sons.